

PERSEC Crime Prediction Model

Comprehensive Risk Factors for Crime Prediction

Based on research in criminology and machine learning, the PERSEC application should incorporate the following categories of variables and factors to create an accurate crime risk prediction model:

1. Spatial and Geographic Factors

Factor	Description	Data Source	Weight
Crime Hotspots	Areas with historically high crime rates	Historical crime data	Very High
Proximity to Points of Interest	Distance to shopping centers, schools, transit stations	POI Data	High
Neighborhood Type	Residential, commercial, industrial, mixed-use	Municipal Zoning Data	High
Population Density	People per square mile/km	Census Data	Medium
Street Network Density	Density and connectivity of streets	Mapbox Data	Medium
Land Use Mix	Diversity of land uses in area	Municipal Zoning Data	Medium
Building Density	Number and density of buildings	Mapbox + Municipal Data	Medium
Street Lighting	Presence and quality of street lighting	Municipal Data + Image Analysis	High
Urban Decay Indicators	Abandoned buildings, graffiti, broken windows	Image Analysis + Reports	High
Visibility	Line of sight, visibility from streets	Calculated from 3D Models	Medium
Public Spaces	Parks, plazas, recreational areas	POI Data	Medium
Natural Surveillance	Eyes on the street, active frontages	Calculated from POI + Image Data	Medium

2. Temporal Patterns

Factor	Description	Data Source	Weight
Time of Day	Hour by hour patterns	System Clock	Very High
Day of Week	Weekday vs weekend patterns	System Calendar	High
Month	Seasonal patterns	System Calendar	Medium
Holidays	Special events and holidays	Holiday Calendar	Medium
School Sessions	School in session vs breaks	School Calendar	Medium
Paydays	Common salary payment days	Financial Calendar	Medium
Special Events	Sports events, concerts, etc.	Events API	Medium
Business Hours	Operating hours of nearby businesses	Business Data	Medium

3. Demographic Factors

Factor	Description	Data Source	Weight
Income Levels	Average household income	Census Data	High
Education Levels	Educational attainment	Census Data	Medium
Age Distribution	Age groups in area	Census Data	Medium
Housing Stability	Homeownership vs rental rates	Census Data	Medium
Population Mobility	Frequency of moving/turnover	Census Data	Medium
Tourist Areas	Areas with high tourist traffic	Tourism Data	Medium
Economic Indicators	Unemployment rates, poverty levels	Economic Data	High
Social Vulnerability Index	Measure of community vulnerability	Census-derived	Medium

4. Environmental Factors

Factor	Description	Data Source	Weight
Weather Conditions	Temperature, precipitation, etc.	Weather API	Medium
Visibility Conditions	Fog, darkness, etc.	Weather API + Calculated	Medium
Lighting Conditions	Natural and artificial lighting	Time + Weather Data	High
Seasonal Effects	Seasonal variations	Calculated from Historical Data	Medium
Noise Levels	Ambient noise	Environmental Data	Low
Air Quality	Pollution levels	Environmental Data	Low

5. Behavioral and Activity Patterns

Factor	Description	Data Source	Weight
Foot Traffic	Pedestrian density	Mobility Data	High
Vehicular Traffic	Vehicle density	Traffic API	Medium
Public Transit Usage	Transit ridership	Transit API	Medium
Nightlife Activity	Bars, clubs, restaurants open late	POI + Business Hours Data	High
Retail Activity	Shopping centers, malls	POI + Business Data	Medium
Social Media Activity	Geotagged posts, check-ins	Aggregated Social Media Data	Low
Mobile Phone Activity	Anonymized device concentration	Telecom Data	Medium

6. Crime-specific Factors

Factor	Description	Data Source	Weight
Previous Crime Types	Types of crimes in area	Crime Database	Very High
Reported Incidents	Non-crime incidents, calls for service	Police Records	High
Crime Clearance Rates	Rate at which crimes are solved	Police Records	Medium
Police Presence	Patrol frequency, proximity to stations	Police Data	High
Security Measures	CCTV, private security	Municipal + Business Data	Medium
Community Programs	Neighborhood watch, outreach	Community Data	Medium

Risk Score Calculation Methodology

The PERSEC application will use a sophisticated machine learning approach to calculate crime risk scores:

1. Base Model Architecture

The crime prediction system will use:

- 1. **XGBoost (Primary Model):** For accurate risk prediction with powerful feature importance capabilities
- 2. **Deep Neural Network (Secondary Model):** For capturing complex spatial and temporal relationships
- 3. **Shapley Additive Explanations (SHAP):** For interpretable predictions that can be explained to users

2. Risk Score Calculation

Each location will receive a normalized risk score from 0-100:

Crime Risk Score = Base Risk Score × Temporal Multiplier × Seasonal Multiplier × Event Multiplier

Where:

- **Base Risk Score:** Calculated from static features (spatial, demographic)
- **Temporal Multiplier:** Adjusts risk based on time patterns (hour, day, month)
- **Seasonal Multiplier:** Adjusts for seasonal crime patterns
- **Event Multiplier:** Adjusts for special events or circumstances

3. Risk Categories

Risk scores will be mapped to five categories:

Risk Score Range	Risk Level	Color Code	User Alert Level
0-20	Minimal Risk	Green	None
21-40	Low Risk	Blue	Informational
41-60	Moderate Risk	Yellow	Awareness
61-80	High Risk	Orange	Caution
81-100	Severe Risk	Red	Warning

Implementation Strategy

1. Data Collection and Preprocessing

1. Crime Data Acquisition:

- Partner with local police departments for historical crime data
- Access public crime databases where available
- Implement privacy protections and data anonymization

2. Feature Engineering:

- Create spatial aggregations at multiple scales (50m, 100m, 250m grids)
- Generate temporal features with cyclic encoding
- Develop neighborhood-level socioeconomic indicators

3. Data Integration:

- Combine crime data with environmental and demographic factors
- Normalize and standardize data from different sources
- Address missing values using imputation techniques

2. Model Training

1. Training Pipeline:

- Split data into training, validation, and test sets with temporal holdout
- Implement grid-based cross-validation to account for spatial autocorrelation
- Train multiple models for different crime types and timeframes

2. Hyperparameter Optimization:

- Use Bayesian optimization for hyperparameter tuning
- Optimize for precision and recall, with emphasis on recall for serious crimes
- Validate on most recent timeframe to test temporal generalization

3. Model Evaluation:

- Use precision, recall, F1-score, and area under ROC curve metrics
- Evaluate spatial accuracy using hit rate and predictive accuracy index
- Assess calibration of probability estimates

3. Real-time Prediction System

1. Prediction API:

- Develop REST API endpoints for real-time crime risk queries
- Implement efficient spatial indexing for fast geographic queries
- Create caching mechanisms for frequently accessed areas

2. Update Frequency:

- Daily refreshes of the base model with new crime data
- Hourly updates for temporal factors
- Real-time updates for special events and emergencies

3. Privacy Protections:

- Ensure all predictions are anonymous and aggregated
- Implement differential privacy techniques
- Avoid predictions that could lead to discriminatory practices

4. Model Explainability

1. Global Interpretability:

- Generate feature importance rankings for the overall model
- Create partial dependence plots for key variables

- Develop summaries of highest-impact factors

2. Local Interpretability:

- Provide SHAP explanations for individual predictions
- Show contributions of top factors to specific risk scores
- Create area-specific risk factor explanations

Technical Implementation with Azure and Mapbox

1. Azure Components

- **Azure Cosmos DB:** Store crime data and risk scores
- **Azure Machine Learning:** Train and deploy prediction models
- **Azure Functions:** Process real-time risk queries
- **Azure Synapse Analytics:** Perform large-scale data analytics
- **Azure Cognitive Services:** Process visual data for environmental factors

2. Mapbox Integration

- **Custom Styling:** Display crime risk heat maps
- **Tilesets:** Create custom tilesets for risk visualization
- **Geocoding API:** Convert addresses to/from coordinates
- **Directions API:** Calculate routes with safety prioritization
- **Vision SDK:** Process street-level imagery for environmental factors

3. Mobile Implementation

The mobile app will implement:

- Real-time crime risk display on map
- Risk-aware route planning
- Contextual risk alerts based on location
- Historical crime pattern visualization
- Customizable risk thresholds and alert preferences

Crime Type Classification

The model will predict risk for specific crime categories:

1. Violent Crimes:

- Assault
- Robbery
- Battery
- Homicide

2. **Property Crimes:**

- Theft
- Burglary
- Motor vehicle theft
- Vandalism

3. **Public Order Crimes:**

- Drug offenses
- Public intoxication
- Disorderly conduct

4. **Financial Crimes:**

- Fraud
- Identity theft
- Forgery

Model Performance Benchmarks

Based on research findings, the model aims to achieve:

- Overall prediction accuracy: 80-85%
- Precision for violent crimes: 75-80%
- Recall for violent crimes: 80-85%
- Area under ROC curve: 0.80-0.85
- Hit rate (percentage of crimes predicted in hotspots): 40-50%

Risk Communication to Users

The PERSEC app will communicate crime risk information to users through:

1. **Map Visualization:** Heat map showing risk levels by color
2. **Route Risk Indicators:** Color-coded segments along navigation routes
3. **Risk Alerts:** Push notifications for entering high-risk areas

4. **Risk Details:** Tap-to-view information about specific risk factors
5. **Safety Recommendations:** Contextual advice based on risk factors

Ethical Considerations

The crime prediction model implementation will consider:

1. **Bias Mitigation:** Techniques to identify and reduce potential bias in predictions
2. **Transparency:** Clear communication about how predictions are made
3. **Privacy Protection:** Ensuring individual privacy is maintained
4. **Fairness:** Avoiding reinforcement of existing disparities in policing
5. **Community Engagement:** Involving communities in model development and deployment

References and Research Basis

This model integrates findings from multiple research studies on crime prediction, including machine learning approaches like XGBoost, deep neural networks, and explainable AI techniques. Key influences include research on spatio-temporal crime patterns, environmental criminology, and modern approaches to interpretable machine learning in public safety applications.